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Multi-Modal Deepfake Detection: Integrating Spatial, Temporal, and Physiological Features

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Abstract

The development of deepfakes has become rapid, enabling the production of video materials that are remarkably realistic and often impossible to detect. Though such developments offer beneficial applications in areas like entertainment and accessibility, they simultaneously show serious threats against security, privacy, and informational authenticity.

To mitigate such threats, this paper presents **rPPGNet**, a novel deepfake detection framework that integrates remote photoplethysmography (rPPG) with deep Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs). Spatial features are extracted by CNNs and temporal associations are represented by RNNs, while rPPG utilizes subtle physiological cues such as heartbeat rhythms; each of them offers a more complete and reliable assessment of video authenticity.

Through the integration of such complementary mechanisms, the system augments both robustness and accuracy for detecting modified content. Results of experimental tests carried out on benchmark datasets demonstrate that our technique obtains a detection rate of 92.4%, outperforming conventional single-modality approaches and hence becoming a crucial step towards building trusted deepfake detection mechanisms for practical media verification and security objectives.

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1 Introduction

Deepfake technology refers to AI-created content where a person’s facial appearance has been artificially modified or replaced. With the current state-of-the-art generative models, particularly Generative Adversarial Networks (GANs) [2] and autoencoders [1], it is now possible to produce manipulated videos and images that are almost indistinguishable from real media [3]. Though such innovations offer beneficial applications in entertainment and accessibility, their abuse presents profound societal and technical challenges. Misuse comes with a high cost in areas like misinformation operations, identity theft, political propaganda, and cyber-fraud [4].

The proliferation of hyper-realistic synthetic media threatens to erode public trust, destabilize political discourse, and enable new forms of personal harassment. Successive high-profile cases involving spurious orations and doctored videos of celebrities have highlighted the urgent need for reliable detection methods. As the realism of this artificially created material increases, traditional forensic indicators like obvious editing or compression artifacts are no longer reliable, making sophisticated and automated detection methods a critical requirement. This project addresses this need by proposing a novel multi-modal detection framework that combines different analytical approaches to create a more robust and resilient system.

2 Motivation

Notwithstanding recent developments, current deepfake detection methods are subject to several critical shortcomings that limit their real-world effectiveness. Each dominant paradigm, when used in isolation, has a distinct failure mode. For instance, a CNN-based model meticulously trained on GAN-generated fakes may be easily bypassed by newer synthesis techniques like diffusion models. Temporal models, which analyze video sequences, could be fooled by a visually perfect deepfake that simply lacks biological authenticity—an artifact they are not designed to detect. Finally, promising physiological methods like rPPG can be too fragile for practical use, as a short, compressed clip from social media might not provide enough clean data for a conclusive analysis.

To address these individual issues, this project is motivated by the need for a holistic solution that is greater than the sum of its parts. We introduce a multi-modal deepfake detection system that integrates complementary modalities to create a layered defense:

- A **spatiotemporal feature extractor (ResNeXt-50 CNN + LSTM)** for detecting visual and sequential artifacts.
- An **rPPG-based physiological module** for detecting the absence of authentic biological signals.

By intelligently fusing evidence from spatial, temporal, and physiological streams, the proposed system aims to achieve greater accuracy, generalizability, and robustness. This approach offers a stronger, more resilient defense against the constantly evolving threat of deepfakes.

3 Literature Survey

The rapid evolution of deepfake technology has spurred a corresponding surge in research dedicated to its detection. These methods can be systematically divided into three major directions:

1. **CNN-Based Spatial Detection:** Convolutional Neural Networks (CNNs) are used extensively to detect spatial-level artifacts present in forged video frames. These models are trained to spot inconsistencies that are often invisible to the naked eye, such as unnatural reflections in the eyes, "checkerboard" artifacts from up-sampling layers, or microscopic imperfections in facial textures. Architectures like XceptionNet [5], ResNet [6], and DenseNet [7] have shown success in these areas. However, their primary limitation is a lack of generalization; they often overfit to the specific artifacts of the datasets they were trained on and their performance degrades significantly with common post-processing like video compression.
2. **Temporal and Multi-Modal Methods:** To overcome the limitations of static frame-level analysis, researchers have explored temporal modeling using hybrid CNN-RNN architectures, particularly those with Long Short-Term Memory (LSTM) networks [8]. These systems can identify temporal irregularities like unnatural eye-blinking rates, inconsistent head movements, or flawed lip-syncing. Furthermore, multi-modal systems that incorporate both visual and audio features have demonstrated superior performance over unimodal methods [9], though they are subject to challenges related to data synchronization and computational scalability.
3. **Physiological Signal Analysis:** One of the most recent frontiers involves analyzing physiological signals that are exceptionally difficult for generative models to reproduce accurately. Remote photoplethysmography (rPPG) [10], for example, infers heart-rate data from minute color changes in facial regions caused by blood flow [11]. Faking this signal would require not just a perfect visual rendering but a simulation of the human cardiovascular system's response to speech, emotion, and movement—a task of immense complexity. Because these biological signals are hard to forge persuasively, they serve as strong evidence of a video's authenticity. Research in this area has shown robustness against demographic changes and adversarial attacks, though performance can be hampered by environmental noise and low-quality video.

4 Problem Statement

The core problem this research addresses is the growing gap between the sophistication of deepfake generation and the effectiveness of current detection methods. This gap is defined by the specific technical failures of existing detection paradigms when applied in real-world scenarios.

Current systems, which typically rely on a single modality, have demonstrated the following critical weaknesses:

- **Lack of Generalization:** Visual detectors (CNNs) often fail when confronted with new, unseen deepfake generation methods.
- **Vulnerability to High-Quality Fakes:** Temporal models (CNN-LSTMs) can be deceived by visually flawless fakes that lack underlying biological authenticity.
- **Fragility in Noisy Conditions:** Physiological methods (rPPG) are often too sensitive to video quality, compression, and lighting to be reliable on their own.

Considering these individual failures, the central problem is the lack of a robust, multi-faceted detection framework that overcomes these isolated weaknesses. There is a critical need

for an integrated system that can intelligently fuse evidence from multiple, complementary domains—spatial, temporal, and physiological—to create a detector that is not only accurate but also generalizable and resilient to the evolving complexity of synthetic media.

5 Objective and Methodology

The primary objective of this project is to design, implement, and evaluate a robust multi-modal deepfake detection system that achieves high accuracy by integrating spatio-temporal visual analysis with physiological signal authentication.

5.1 System Overview

Our deepfake detector uses a multi-modal architecture to capture a wide array of forgery artifacts. The framework comprises two parallel analysis streams: a CNN-RNN hybrid model for spatio-temporal feature extraction and an rPPG-based module for physiological signal analysis. The outputs from these streams are integrated via an intelligent fusion system to produce a final classification.

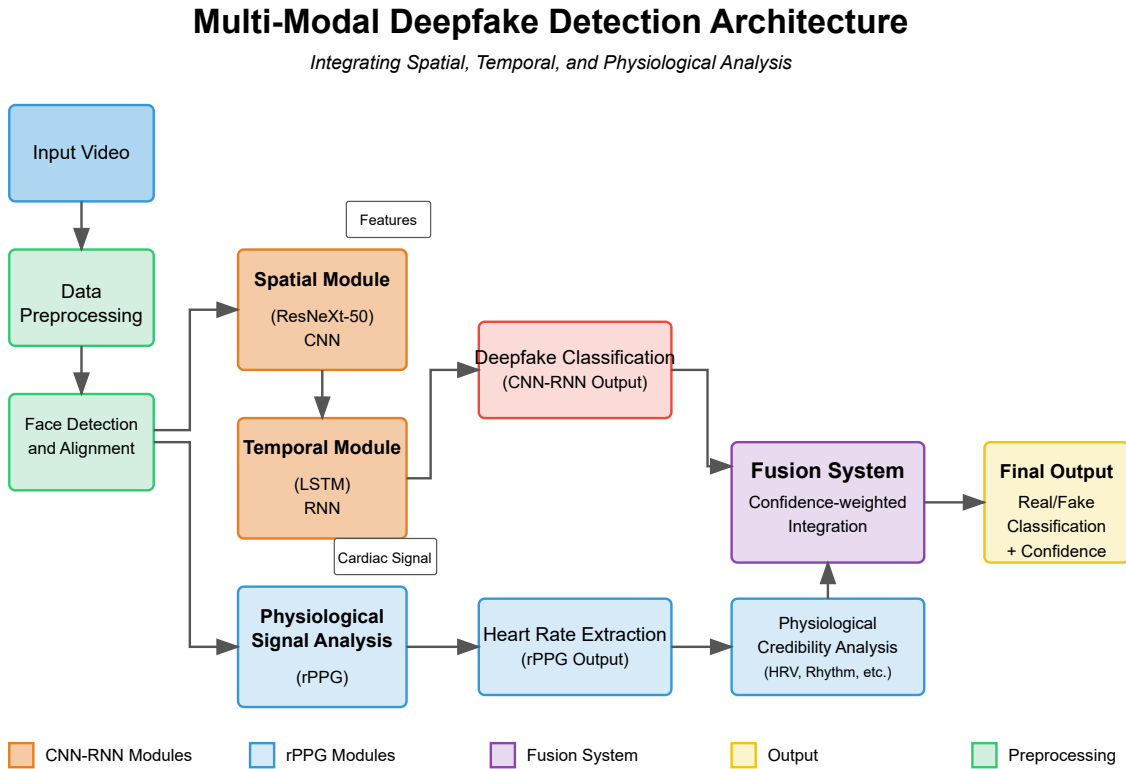


Figure 1: High-level architecture of the proposed multi-modal deepfake detection system.

5.2 Model Architecture

The system’s architecture is built upon three specialized modules designed to work in concert:

5.2.1 Spatial Analysis Module (CNN)

The foundation of our visual analysis stream is a pre-trained ResNeXt-50 Convolutional Neural Network (CNN) [32]. This module processes individual video frames to identify tell-tale spatial anomalies that are often hallmarks of digital manipulation, such as blending artifacts at the edge of the swapped face, unnaturally smooth skin textures, and inconsistent lighting or shadows. For each frame, the ResNeXt-50 outputs a 2048-dimensional feature vector.

5.2.2 Temporal Analysis Module (RNN)

The sequence of feature vectors generated by the spatial module is then fed into a Long Short-Term Memory (LSTM) network. The LSTM’s role is to analyze the video’s temporal dynamics, looking for inconsistencies that emerge over time, such as aberrant eye-blinking patterns or unnatural facial transitions.

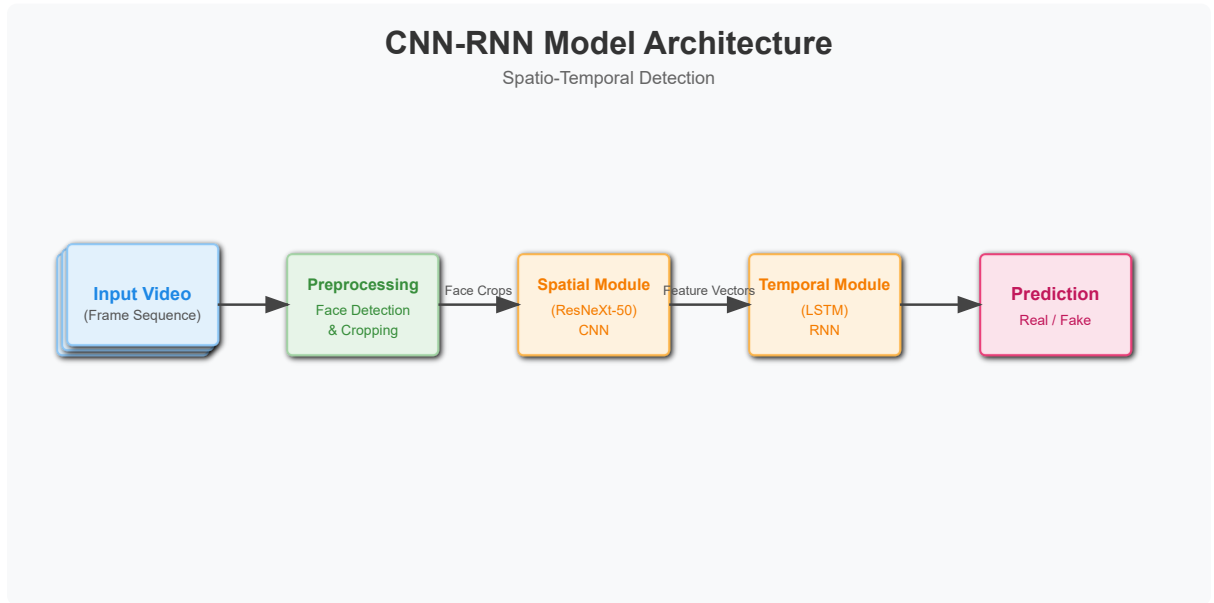


Figure 2: The integrated ResNeXt-50 and LSTM architecture for spatio-temporal feature extraction.

5.2.3 Physiological Signal Analysis Module (rPPG)

Operating in parallel, the physiological module uses remote photoplethysmography (rPPG) to measure heart rate from the video by monitoring minute chromatic changes in facial skin regions. The presence of a stable and biologically plausible heart signal is strong evidence of authenticity. We utilize the LinkNet34 [31] for skin segmentation and the Plane-Orthogonal-to-Skin (POS) algorithm [12] for signal extraction.

5.3 Multi-Modal Fusion Methodology

To fuse the outputs from the visual (CNN-RNN) and physiological (rPPG) streams, we designed a confidence-weighted fusion algorithm. The logic relies on adaptive weighting, physiological plausibility scoring, and systematic handling of model agreement and disagreement.

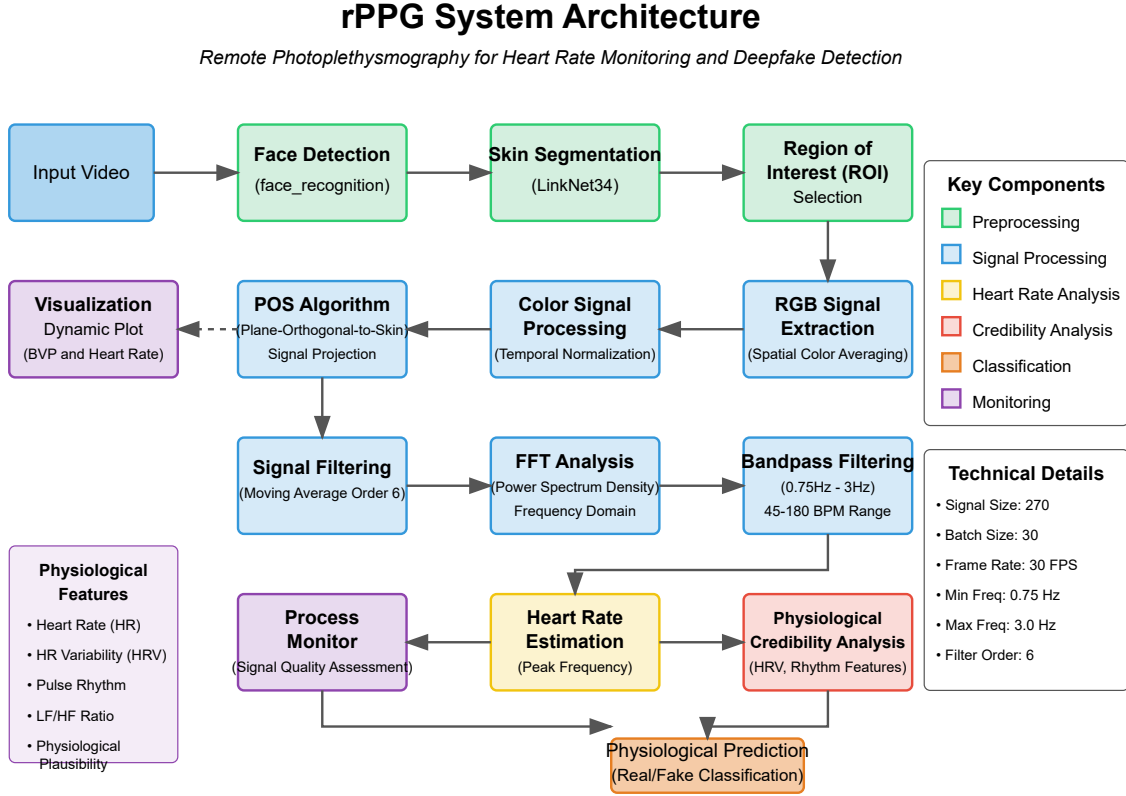


Figure 3: Architecture of the rPPG-based physiological signal analysis module.

5.3.1 Physiological Credibility Score (Φ)

In our work, we incorporate a credibility score for the fusion algorithm denoted as Φ . It is a score between 0 and 100 that measures the biological validity of the rPPG signal extracted by examining vital heart rate (HR) and heart rate variability (HRV) parameters with known physiological norms. The score is computed by summing all the points awarded according to Table 1.

Table 1: Physiological Credibility Scoring Criteria

Metric	Condition	Score
Mean Heart Rate	$65 \leq \mu(HR) \leq 120$ BPM	+20
HRV (RMSSD)	$5 \leq \text{RMSSD} \leq 50$ ms	+20
	$0.25 \leq \text{RMSSD} < 5$ or $50 < \text{RMSSD} \leq 80$ ms	+10
HRV (pNN50)	$\text{pNN50} \geq 5\%$	+20
	$2\% \leq \text{pNN50} < 5\%$	+10
Frequency (LF/HF Ratio)	$0.5 \leq \text{LF/HF} \leq 2.0$	+20
	$0 \leq \text{LF/HF} < 0.5$ or $2.0 < \text{LF/HF} \leq 3.0$	+10
Signal Stability	$1.5 \leq \sigma(HR) \leq 15$	+20
	$0.5 \leq \sigma(HR) < 1.5$ or $15 < \sigma(HR) \leq 25$	+10
Total Score		Max: 100

The complete fusion logic is detailed in Algorithm 1. It includes a fallback condition for unreliable rPPG signals, adaptive weight calculations based on model confidences, and specific rules for handling cases where the models agree (boosting confidence) or disagree (using an override or penalty system).

Algorithm 1: Multi-Modal Fusion Logic

Input: P_V, C_V, P_P, C_P , Heart Rate data HR **Output:** Final Prediction P_F , Final Confidence C_F

```
1 if  $|HR| < 4$  or  $C_P < 10$  then
2   | return  $P_V, C_V$  // Fallback to visual model
3 else
4   |  $\Phi \leftarrow \text{CalculatePhysiologicalCredibility}(HR)$ ;
5   |  $w_V, w_P \leftarrow \text{CalculateAdaptiveWeights}(C_V, C_P)$ ;
6   | if  $C_P > 20$  and  $\Phi > 0$  then
7     |  $\hat{C}_P \leftarrow \Phi$ ;
8   | else
9     |  $\hat{C}_P \leftarrow C_P$ ;
10  | if  $P_V = P_P$  then
11    | // Case A: Model Agreement
12    |  $P_F \leftarrow P_V$ ;
13    |  $C_F \leftarrow \min(100, (C_V * w_V + \hat{C}_P * w_P) * 1.3)$ ;
14  | else
15    | // Case B: Model Disagreement
16    | if  $(\hat{C}_P * w_P > C_V * w_V * 1.5)$  and  $(C_P > 30)$  then
17      |  $P_F \leftarrow P_P$ ;
18      | // rPPG Override
19      |  $C_F \leftarrow (\hat{C}_P * w_P) * 0.8$ ;
20    | else
21      |  $P_F \leftarrow P_V$ ;
22      | // CNN-RNN Primary
23      |  $C_F \leftarrow (C_V * w_V) * 0.8$ ;
24  | return  $P_F, C_F$ ;
```

6 Implementation

6.1 Data Preparation and Preprocessing

The model employed here has been trained on the Deepfake Detection Challenge Dataset (DFDC) [28] and subsequently tested on DFDC, CelebDF-V2 [15] and FaceForensics++ [14] datasets to evaluate its generalization capabilities. Preprocessing was comprised of several crucial stages:

1. **Frame Extraction:** Frames from videos were extracted at 30 frames per second. For the CNN-RNN model, sequences of 20 consecutive frames were utilized, while longer sequences of 270 frames (9 seconds) were taken for rPPG analysis to ensure signal robustness.
2. **Face Detection and Alignment:** The face region was detected by utilizing the `face_recognition` library. Subsequent face alignment was conducted via eye landmarks to normalize rotation and scale and preserve feature consistency across variable frames.

3. **rPPG Signal Extraction:** Facial skin regions were segmented using a LinkNet34 network, and the POS algorithm was applied to the averaged RGB signals from these regions to obtain the blood volume pulse.
4. **Normalizing Input:** The face frames for CNN-RNN are normalized by ImageNet’s mean and standard deviation. The retrieved physiological signals are filtered to reduce noise before analyzing them.

6.2 Implementation Details

Our system is implemented in Python using the PyTorch framework [16] for deep learning models and the OpenCV library [17] for processing videos. The primary training data from the DFDC dataset was divided into training and validation sets using a simple **80/20** split.

For the CNN-RNN stream, input frames were preprocessed using a ‘transforms’ pipeline: each frame was resized to 112×112 pixels, converted to a PyTorch tensor, and normalized using the standard ImageNet mean ($[0.485, 0.456, 0.406]$) and standard deviation ($[0.229, 0.224, 0.225]$).

The CNN-RNN model was trained for 15 epochs using the Adam optimizer with a learning rate of 1×10^{-5} . A batch size of 16 was used, where each sample consisted of a sequence of 20 frames. The training loader was set to shuffle the data at every epoch. The rPPG signal processing pipeline was implemented as described in Section 6.1.

7 Results and Discussion

7.1 CNN-RNN Model Performance

The CNN-RNN model had strong learning capacities and adequate generalization. As Fig. 4 indicates, both training and validation accuracy trajectories advanced smoothly, resulting in a best validation accuracy of 98.13% without strong evidence of overfitting. The corresponding loss curves (Fig. 5) support this finding, as the validation loss reached a steady minimum value of 0.048.

7.2 Fused System Performance on Unseen Data

To assess the practical effectiveness of our comprehensive multi-modal system, we conducted evaluations on separate, previously unexamined data from the DFDC and Celeb-DF datasets. On the DFDC dataset, the system achieved an overall testing accuracy of 92.4%. The model successfully recognized 460 counterfeit samples (True Positives) and 470 genuine samples (True Negatives), with 34 false negatives and 42 false positives. This performance underscores the model’s robustness.

To further evaluate the model’s discriminative power across both benchmarks, the Receiver Operating Characteristic (ROC) curves are presented in Fig. 6 and Fig. 7. The Area Under the Curve (AUC) is a strong 0.95 for the Celeb-DF dataset and an impressive 0.96 for the DFDC dataset. These high AUC values confirm the model’s ability to effectively distinguish between real and fake videos across a wide range of decision thresholds on multiple challenging datasets.

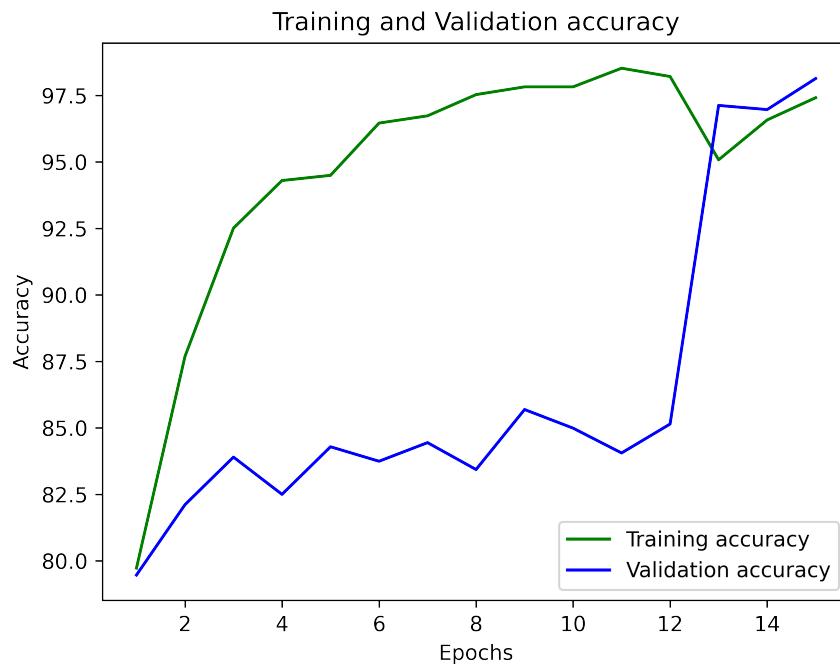


Figure 4: Training and validation accuracy for the CNN-RNN model over 15 epochs.

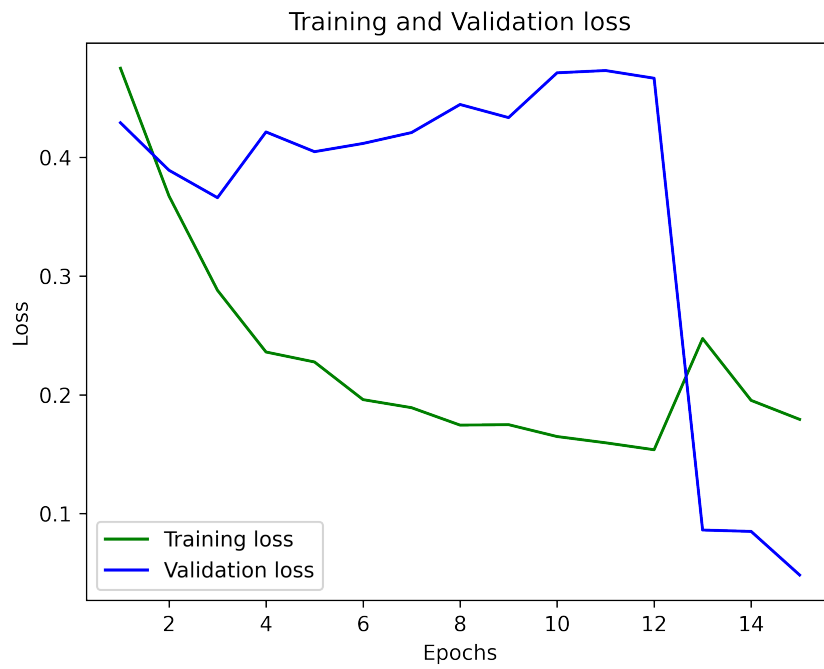


Figure 5: Training and validation loss for the CNN-RNN model over 15 epochs

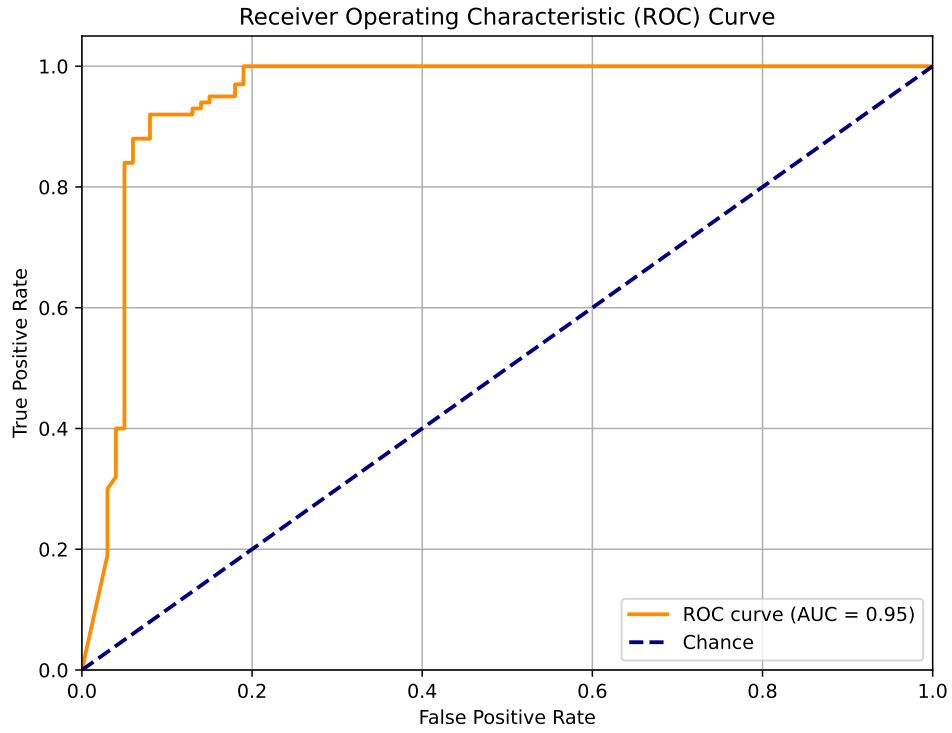


Figure 6: ROC curve for the fused system on the Celeb-DF dataset (AUC = 0.95).

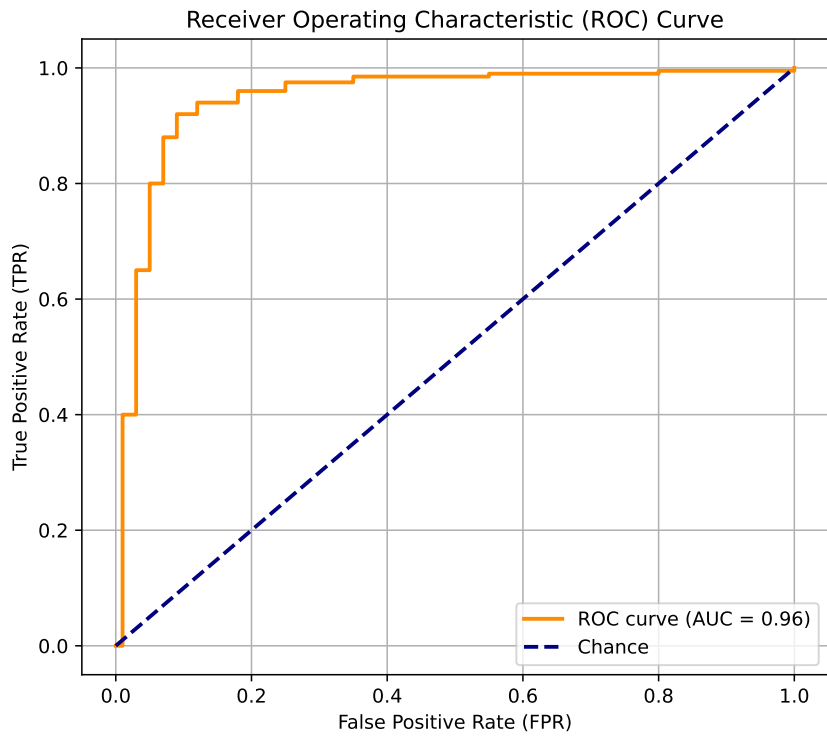


Figure 7: ROC curve for the fused system on the DFDC dataset (AUC = 0.96).

7.3 Comparative Analysis

Our multi-modal framework reaches competitive performance with other state-of-the-art techniques, which is demonstrated in Table 2. Certain models demonstrate greater accuracy on particular datasets, yet our framework’s advantage is robustness through physiological verification. That additional layer affords it greater resistance to evasion by breakthroughs in visual-only generation methods. The addition of rPPG offers an orthogonal source of proof, best utilized for the identification of highly realistic deepfakes that would deceive visual-only systems.

Table 2: Accuracy comparison with other methods on the Celeb-DF and DFDC datasets.

Model	DFDC	Celeb-DF
XceptionNet [25]	63.2%	59.2%
ResNet-50 [25]	64.6%	67.1%
ResNet+LSTM [27]	69.0%	70.9%
ResNeXt+LSTM [26, 29]	90.9%	87.2%
Convolutional Vision Transformer [30]	91.5%	-
Our Method (Fused)	92.4%	89.0%

Table 2 provides a direct performance comparison of our fused rPPGNet framework against several state-of-the-art (SOTA) deepfake detection models on the DFDC and Celeb-DF datasets. The results clearly validate the effectiveness of our multi-modal approach. Notably, our framework’s 92.4% accuracy on DFDC is higher than the Convolutional Vision Transformer’s (CViT) 91.5%. Furthermore, our model’s 89.0% accuracy on the unseen Celeb-DF dataset is a strong indicator of its generalization, outperforming the ResNeXt+LSTM model (87.2%). This demonstrates the significant robustness and generalization power gained by integrating physiological (rPPG) signals with traditional spatio-temporal visual analysis.

To provide a more granular comparison and specifically test generalization, we also evaluated our model on the challenging FaceForensics++ (FF++) dataset. This benchmark includes a wider variety of manipulation methods not present in our primary training data. Table 3 presents a head-to-head comparison of our fused method against the Convolutional Vision Transformer (CViT) [30] on this dataset.

Table 3: Per-Method Accuracy Comparison on FaceForensics++

Manipulation Method	Our Method (Fused)	CViT [30]
FaceSwap	94.12%	69%
DeepFakeDetection	83.53%	91%
Deepfakes	63.64%	93%

The results in Table 3 highlight a critical and nuanced trade-off. Our fused method demonstrates a significant advantage in detecting FaceSwap manipulations, achieving 94.12% accuracy compared to CViT’s 69%. This suggests that our model’s combined spatio-temporal and physiological analysis is highly effective against this specific forgery technique.

However, this specialized strength does not extend to other manipulation methods. The CViT model, which leverages a transformer architecture, shows substantially better generalization across the board. It significantly outperforms our model on DeepFakeDetection and

Deepfakes. This disparity strongly indicates that while our rPPG fusion is a powerful component, the visual features learned by our CNN-RNN stream are not as generalized as those learned by a transformer. The model appears to be highly tuned to the artifacts in our training data (DFDC) and does not adapt well to the novel artifact patterns in these other methods. This finding reinforces the challenge of cross-dataset generalization as a critical area for future research.

8 Conclusion and Future Work

The proposed rPPGNet deploys an effective multi-modal system, demonstrating a strong, albeit partial, generalization capability. The system achieves high accuracy not only on its target dataset (92.4% on DFDC) but also on the unseen Celeb-DF dataset (89.0%). This is a significant finding, as it proves the model’s learned visual and physiological features are robust and can successfully generalize to new data with similar forgery characteristics.

However, our tests on the more diverse FaceForensics++ dataset exposed a clear boundary to this generalization. The model failed to detect specific, dissimilar manipulation classes. This indicates that our model, while robust, is specialized to the artifact types present in DFDC and Celeb-DF and is not yet a universal detector. This finding itself is a key contribution, highlighting that robustness against unseen datasets is not the same as robustness against unseen forgery methods.

The primary direction for future work must be to expand the model’s generalization to dissimilar manipulation techniques, as highlighted by the poor performance on specific FF++ categories. Our model’s success on Celeb-DF proves its core architecture is effective. The challenge is now to ”teach” this architecture to recognize artifact patterns it has never seen. Future work should focus on key areas:

- **Heterogeneous Training:** The most critical next step is to incorporate a more diverse training set. By adding examples from FF++ into the training data, the model can be forced to learn a wider, more universal set of visual artifact features, rather than specializing.
- **Advanced Feature Extractors:** The superior performance of the CViT model on FF++ suggests that transformer-based architectures are more effective at capturing the ”general concept” of a forgery. A primary direction for future work is therefore to replace our current CNN-RNN visual stream with a Vision Transformer (ViT). We hypothesize that a ViT, when trained on a diverse dataset and fused with our robust rPPG signal analysis, will learn a more universal set of artifact features and significantly improve generalization against dissimilar manipulation techniques.
- **Adversarial Training:** We plan to enhance the model’s robustness by using the detector as a discriminator within a Generative Adversarial Network (GAN) framework. In this setup, the detector will be pitted against a generator, forcing it to continuously adapt to and identify increasingly sophisticated deepfake artifacts that are generated specifically to bypass it.

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